East West University

Department of MDS

MAT 205

Section: 8

Name: ....

Marks Obtained

Date: 17.08.2026

Quiz No.: 03

ID. No.: ....

Topics: Vector Space and Complex Analysis

Total Marks: 20 | Time: 25 minutes

1. Write the first fundamental theorem of subspace. Express the vector $\underline{v}$ as a linear combination of the vectors $\underline{v_1}, \underline{v_2}$, where
   $\underline{v} = (2, -5, 4)$ and $\underline{v_1} = (1, -3, 2), \underline{v_2} = (2, -1, 1)$. **5**
2. Verify whether the function $f(z) = \frac{1}{z}; z \neq 0$ is analytic. Is it entire? Justify your answer. **6**
3. For the function $f(z)$, locate and name all the singularities in the finite z-plane, where
   $f(z) = \frac{z^8+z+2}{(z-1)^3(3z+2)^2}$. Discuss briefly the essential and removable singularities in relation to the given function. **9**